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## Project Report On

# SIM SWAP FRAUD IN INDIA — 2024

Submitted in partial fulfilment of the requirement for the project work for the degree of

MASTER OF COMPUTER APPLICATION (MCA)

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BANGALORE UNIVERSITY

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## CERTIFICATE

This is to certify that the project synopsis entitled "SIM SWAP FRAUD IN INDIA — 2024" is submitted by Ms.Janani R (USN: U03BX23S0032) in partial fulfilment of the requirements for the award of the degree of Bachelor of Computer Applications (BCA) from Bangalore University for the academic year 2025–2026.

The project synopsis has been prepared under the guidance and supervision of the undersigned. The synopsis is submitted for academic evaluation in accordance with the applicable department and university guidelines.

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# CASE STUDY: SIM SWAP FRAUD IN INDIA — 2024

## 1. Title

**SIM Swap Fraud in India: A Case Study of the ₹7.60 Crore Mumbai Cyber Fraud Incident of 2024**

**Subject:** Cybersecurity / Cyber Crime

**Case Study Type:** Real-World Cybersecurity Incident

**Year:** 2024

**Location:** Mumbai, Maharashtra, India

# 2. Abstract

SIM swap fraud is a form of cyber-enabled financial crime in which criminals fraudulently obtain control of a victim's mobile number by getting a replacement SIM or eSIM activated. Once control of the number is obtained, criminals may receive SMS-based authentication messages and banking alerts intended for the genuine customer. This makes SIM swap attacks particularly dangerous when mobile numbers are closely connected to online banking and other digital services.

A significant example occurred in Mumbai in December 2024, when a business family discovered that their mobile numbers had stopped functioning normally. The telecom service provider informed them that replacement eSIMs had been issued. Subsequent checking revealed that approximately ₹7.60 crore had been transferred from company accounts. The victims immediately contacted the cybercrime helpline 1930, following which cyber police and financial institutions worked to freeze relevant accounts and approximately ₹4.65 crore was secured. (The Indian Express)

This case demonstrates how weaknesses involving identity verification, SIM replacement, social engineering and dependence on SMS-based authentication can combine to create substantial financial risk. It also highlights the importance of rapid reporting, telecom security, banking controls, multi-factor authentication and public awareness.

# 3. Introduction

Mobile phones have become an important component of modern digital banking. Customers frequently use their registered mobile numbers to receive transaction alerts, authentication messages and account-related notifications. As a result, a mobile number can become an important security factor in financial transactions.

SIM swap fraud exploits this relationship between mobile identity and financial services. According to India's National Cyber Crime Reporting Portal, a SIM Swap Scam occurs when criminals fraudulently obtain a replacement SIM for a victim's registered mobile number. The criminal can then receive OTPs and alerts associated with the victim's number. (Cyber Crime Portal)

A typical SIM swap attack can involve several stages. Criminals may first collect information about a target through phishing, vishing, smishing, leaked information or other social-engineering techniques. They may then attempt to convince or manipulate a telecom operator or exploit weaknesses in the SIM replacement process. After a replacement SIM is activated, the genuine customer's SIM becomes inactive and communications are redirected to the attacker's device. Delhi Police describes this general process as involving the collection of personal information, fraudulent SIM replacement and subsequent use of OTPs for financial transactions. (Delhi Police Cyber Crime Unit)

The problem became particularly important in India during 2024 because of the increasing dependence on mobile numbers for digital services. The Department of Telecommunications identified approximately **6.80 lakh suspected mobile connections** obtained using invalid or forged identity/address documents in May 2024 and ordered their re-verification. (Press Information Bureau)

The Mumbai incident examined in this case study demonstrates how SIM-related attacks can cause major financial damage and why cooperation between victims, telecom operators, banks, cybercrime investigators and government agencies is essential.

# 4. Incident Description

## 4.1 Background

The selected incident occurred in **Mumbai in December 2024** and involved a business family operating a steel-trading company.

According to reports, the victims suddenly found that their mobile phones were not reachable even though the devices themselves were switched on. When they approached their mobile service provider, they were informed that replacement SIMs/eSIMs had been issued for their numbers. (The Times of India)

This was an important warning sign because the victims had not intentionally requested such replacement SIMs.

## 4.2 Discovery of the Fraud

After discovering the SIM-related problem, the victims checked their company's bank accounts. They found that approximately **₹7.60 crore had been transferred fraudulently**.

The attackers had effectively taken control of the mobile numbers associated with the victims. This allowed the criminals to receive communications that would normally reach the legitimate customers.

The victims reported the incident to the cybercrime authorities. The national cybercrime helpline **1930** was used to initiate an emergency response. (The Indian Express)

## 4.3 Why the Incident Was Significant

The case was significant because:

- The financial loss was extremely large.
- Multiple mobile numbers were reportedly affected.

- The criminals used SIM replacement/eSIM activation as part of the attack.
- Banking transactions were possible because the attackers gained control of the victim's mobile communications.
- Rapid reporting through 1930 helped authorities take action.
- The incident demonstrated the relationship between telecom security and banking security.

## 5. Attack Type / Vulnerability

The primary attack type was **SIM Swap Fraud**, also called **SIM hijacking** or **SIM replacement fraud**.

The National Cyber Crime Reporting Portal specifically identifies SIM Swap Scam as a cybercrime in which criminals obtain a replacement SIM against a registered mobile number and subsequently receive OTPs and financial alerts. (Cyber Crime Portal)

### Major vulnerabilities involved

**1. Identity verification weakness**: The attacker needs enough information to convince the telecom provider that a replacement SIM request is legitimate.

**2. Social engineering**: Personal information can be obtained through deceptive calls, messages, phishing websites or other forms of manipulation.

**3. Dependence on SMS authentication**: When SMS OTPs are the primary additional authentication mechanism, control of the mobile number can become extremely valuable to an attacker.

**4. Telecom account compromise**: A fraudulent SIM replacement can disconnect the legitimate user while transferring mobile connectivity to another SIM/eSIM.

**5. Banking dependence on registered mobile numbers**: Banking systems frequently use registered mobile numbers for alerts and authentication, making SIM security an important part of financial security.

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# 6. Root Cause Analysis

The root cause should not be viewed as a single technical failure. SIM swap attacks generally result from several weaknesses interacting with one another.

### Root Cause 1: Compromise of personal information

The attacker must generally obtain sufficient information about the victim. This may come from phishing, social engineering, publicly available information, compromised accounts or previously exposed data.

### Root Cause 2: Fraudulent SIM replacement

The key vulnerability is the ability to obtain a replacement SIM using fraudulent identity information or by manipulating the verification process.

## Root Cause 3: Authentication dependency

If financial systems rely heavily on SMS OTPs, control over the registered mobile number can provide attackers with an important authentication channel.

## Root Cause 4: Insufficient transaction controls

Even after obtaining OTPs, extremely large financial transactions should ideally trigger additional risk-based verification.

## Root Cause 5: Delayed recognition

SIM swap attacks can be difficult to notice initially. A customer may interpret sudden loss of mobile connectivity as a normal network problem.

The Government of India subsequently highlighted the importance of stronger KYC procedures and a robust SIM swap/replacement process. (Press Information Bureau)

# 7. Impact Analysis:

**7.1 Financial Impact :** The most direct impact was financial. Approximately **₹7.60 crore** was reported as transferred from the company's accounts. Approximately **₹4.65 crore was secured through the cybercrime response**. (The Indian Express)This shows that cybercrime can affect not only individuals but also businesses with substantial financial resources.

**7.2 Operational Impact:** The victims temporarily lost normal control of their mobile numbers. Mobile connectivity is increasingly important for communication, banking and business operations.

**7.3 Security Impact:** The case demonstrated that compromise of a telecom identity can become a gateway to financial accounts.

**7.4 Psychological Impact:** Victims of large financial cybercrime may experience stress, anxiety and loss of confidence in digital banking.

**7.5 Business Impact:** For a business, unauthorized transfers can affect:

- Working capital
- Supplier payments
- Employee payments
- Customer relationships
- Business continuity
- Reputation

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## 8. Detection & Response

The incident provides several useful examples of detection and response.

**Step 1: Loss of mobile connectivity:** The victims noticed that their mobile numbers were no longer functioning normally.

**Step 2: Telecom verification:** The victims contacted their telecom provider and discovered that replacement eSIMs had been issued.

**Step 3: Bank account verification:** After learning about the SIM replacement, they checked their bank accounts and identified unauthorized transactions.

**Step 4: Reporting to cybercrime authorities:** The victims contacted the **1930 cybercrime helpline**.

**Step 5: Account freezing:** Cybercrime authorities coordinated with financial institutions to identify and freeze accounts associated with the fraudulent transfers.

**Step 6: Recovery:** Approximately ₹4.65 crore was secured/recovered. The Indian government's Citizen Financial Cyber Fraud Reporting and Management System is specifically designed to support immediate reporting and action against financial cyber fraud. (Press Information Bureau)

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## 9. Security Measures & Mitigation

Several security measures can reduce the risk of SIM swap attacks.

**9.1 Strong SIM replacement verification:** Telecom operators should use strong identity verification before activating replacement SIMs.The Department of Telecommunications has introduced stronger KYC requirements and specifically emphasized a robust KYC process for SIM swap/replacement. (Press Information Bureau)

**9.2 SIM replacement alerts:** Customers should receive immediate notification whenever a SIM replacement is requested or completed.

**9.3Transaction monitoring:** Banks should monitor unusual transactions, particularly: Very large transfers, New beneficiaries,Multiple transfers within a short period,Unusual login/device behaviour

**9.4 Multi-factor authentication:** Where available, customers should use stronger authentication methods rather than relying entirely on SMS OTPs.

**9.5 Customer awareness:** Users should understand that telecom problems can sometimes be a security warning.

**9.6 Immediate reporting:** In a financial cyber fraud, victims should immediately contact **1930** and report the incident through the National Cyber Crime Reporting Portal. (Cyber Crime Portal)

**9.7 Telecom-sector monitoring:** The government has also used technology to identify suspicious mobile connections. In May 2024, DoT reported approximately 6.80 lakh connections requiring re-verification because of suspected invalid or forged KYC documentation. (Press Information Bureau)

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## 10. Lessons Learned

The Mumbai incident provides several important cybersecurity lessons.

**Lesson 1: Mobile numbers are security assets:** A mobile number should not be treated simply as a communication facility. It can be connected to banking, email and other important accounts.

**Lesson 2: Loss of network connectivity can be a warning sign:** Unexpected loss of mobile service should be investigated promptly, particularly when it occurs without an obvious technical reason.

**Lesson 3: SMS OTP is not completely risk-free:** SMS-based authentication can be weakened if an attacker gains control of the associated SIM.

**Lesson 4: Speed matters:** The recovery of a substantial portion of the money demonstrates the importance of immediate reporting.

**Lesson 5: Cybersecurity requires cooperation:** Telecom operators, banks, cybercrime investigators and customers must cooperate because the attack crosses multiple systems.

**Lesson 6: KYC controls are critical:** Weaknesses in customer verification can become an entry point for larger cybercrime operations.

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## 11. Recommendations

**For Individuals:** Never share banking credentials, PINs or authentication information with unknown callers.

1. Treat unexpected loss of mobile connectivity seriously.
2. Contact the telecom provider immediately if an unexplained SIM replacement is suspected.
3. Enable stronger authentication methods where available.

4. Monitor bank transactions regularly.
5. Set transaction limits where banking services allow them.
6. Report financial cybercrime immediately through **1930**.

**For Businesses:** Use multiple layers of authentication.

1. Require additional approval for high-value transactions.
2. Establish transaction monitoring systems.
3. Train employees about phishing and social engineering.
4. Maintain emergency cyber-incident response procedures.
5. Separate financial authorization responsibilities among employees.

**For Telecom Operators:** Strengthen SIM replacement verification.

1. Monitor unusual SIM replacement patterns.
2. Maintain detailed audit records.
3. Immediately alert customers about SIM changes.
4. Conduct regular audits of SIM-issuing points of sale.
5. Block or investigate suspicious SIM activation activity.

**For Banks:** Use behavioural transaction monitoring.

1. Apply additional verification to unusual high-value transactions.
2. Detect changes in SIM status where technically and legally feasible.
3. Provide immediate transaction alerts through multiple channels.
4. Improve coordination with cybercrime authorities.

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# 12. Conclusion

The 2024 Mumbai SIM swap fraud demonstrates how a relatively small weakness in telecom identity management can develop into a major financial cybercrime. In the reported incident, criminals obtained control of the victims' mobile numbers through fraudulent SIM/eSIM replacement, after which approximately ₹7.60 crore was transferred from company accounts. The victims' quick response and the involvement of the cybercrime helpline and financial institutions helped secure approximately ₹4.65 crore. (The Indian Express)The case highlights an important cybersecurity principle: **security must be considered across the entire digital ecosystem**. A bank may have strong internal controls, but if an attacker can compromise the customer's telecom identity, some authentication mechanisms can become vulnerable.

India's subsequent efforts to strengthen telecom KYC, identify suspicious mobile connections and improve cyber-fraud reporting show that SIM security is being treated as an important part of the country's broader cybersecurity framework. (Press Information Bureau)Ultimately, preventing SIM swap fraud requires a combination of strong identity verification, secure telecom procedures, robust banking controls, customer awareness and rapid incident response. The Mumbai case also demonstrates that **early detection and immediate reporting can significantly reduce financial losses**.

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THANK YOU

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CYBER SECURITY REPORT

ACADEMIC YEAR 2026-2027